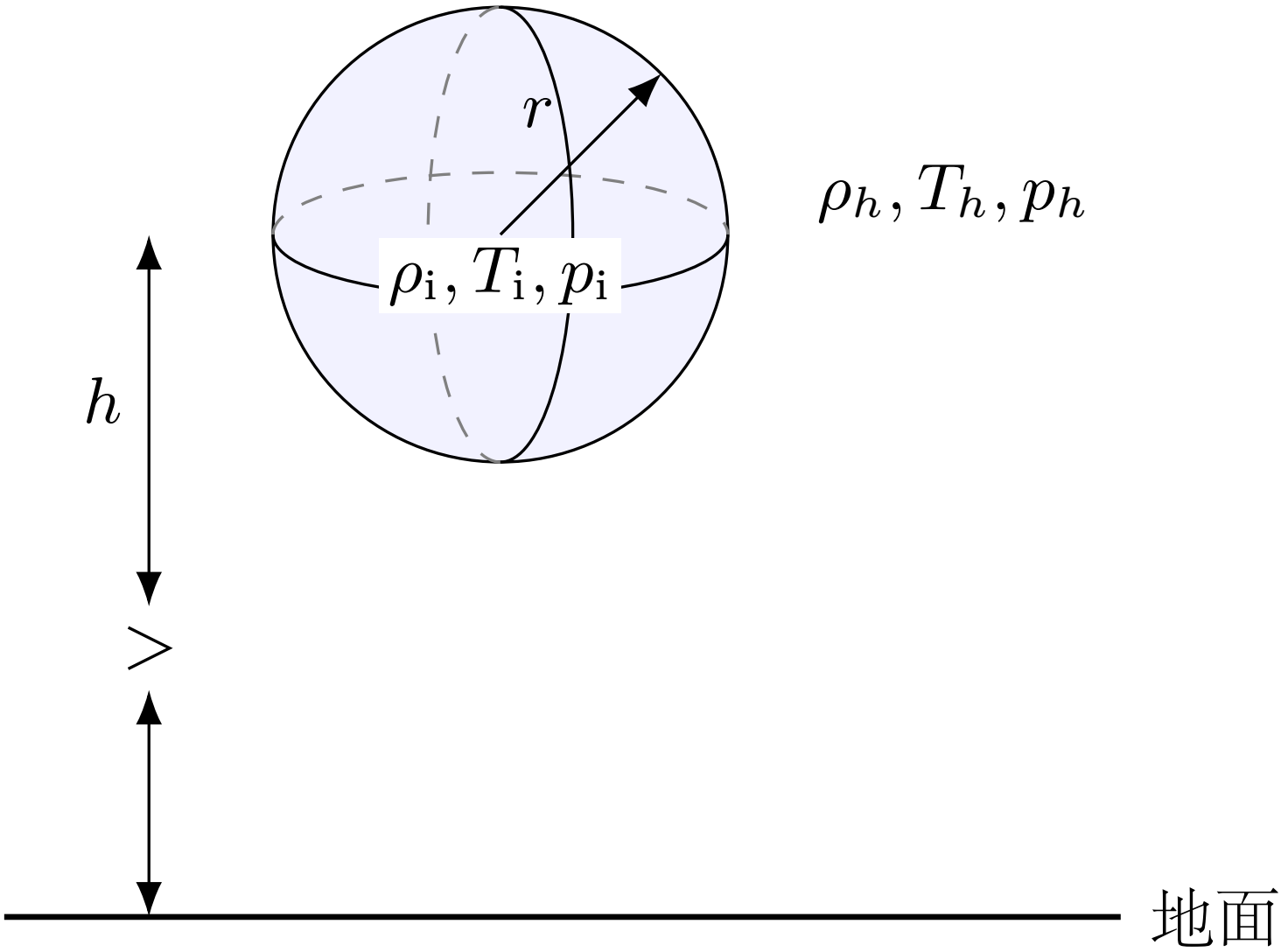
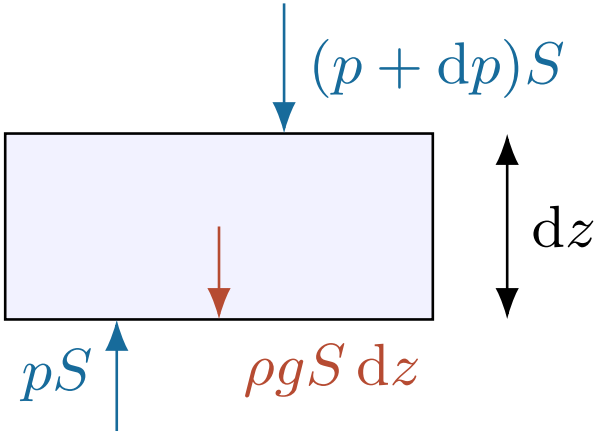
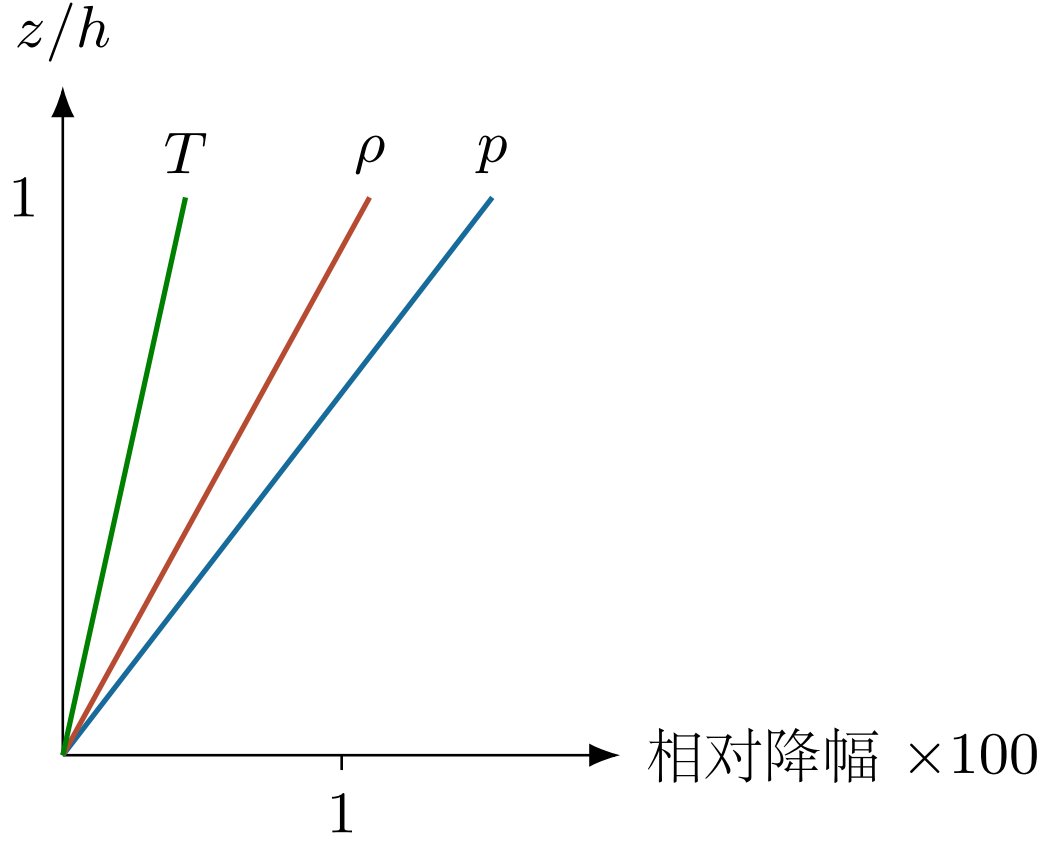
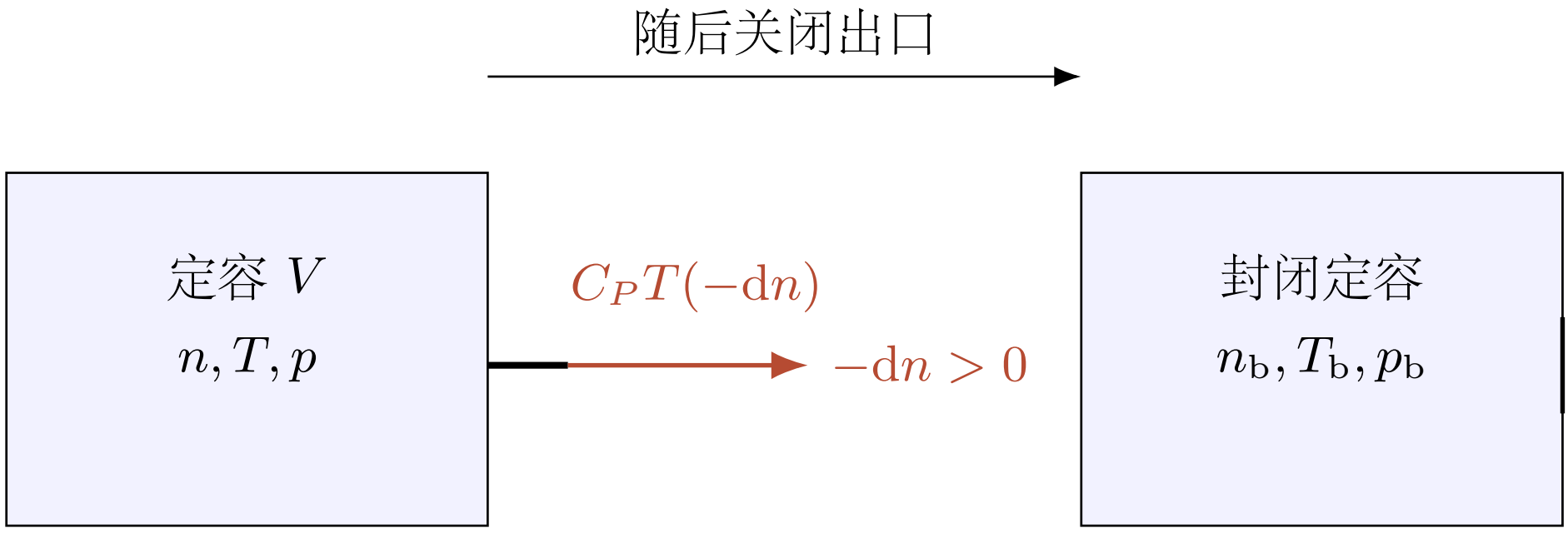


高度轴截断；气囊体积固定

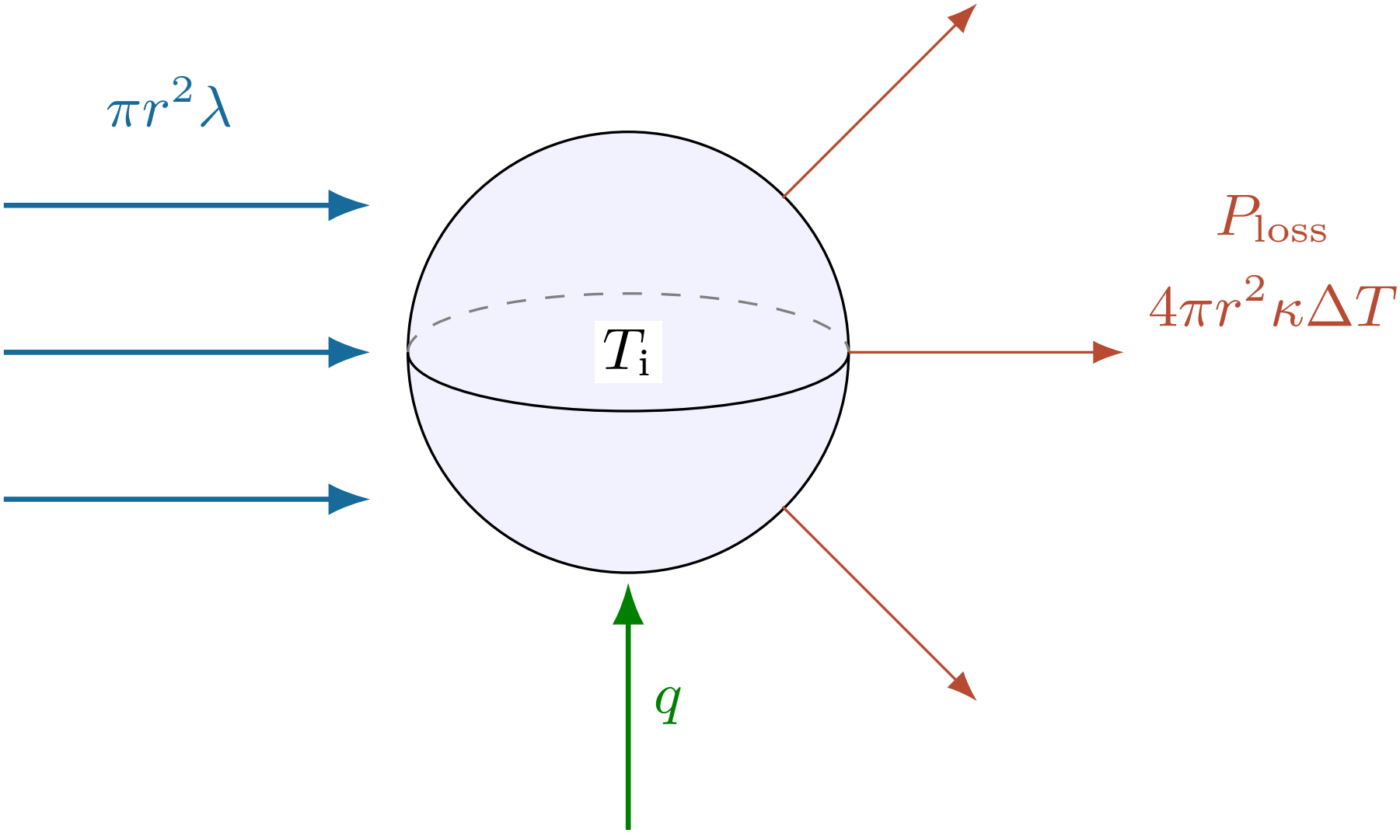

$(p + \mathrm{d}p)S$
$\mathrm{d}z$
pS
$\rho g S\,\mathrm{d}z$


z/h
1
T
ρ
p
1
相对降幅 ×100

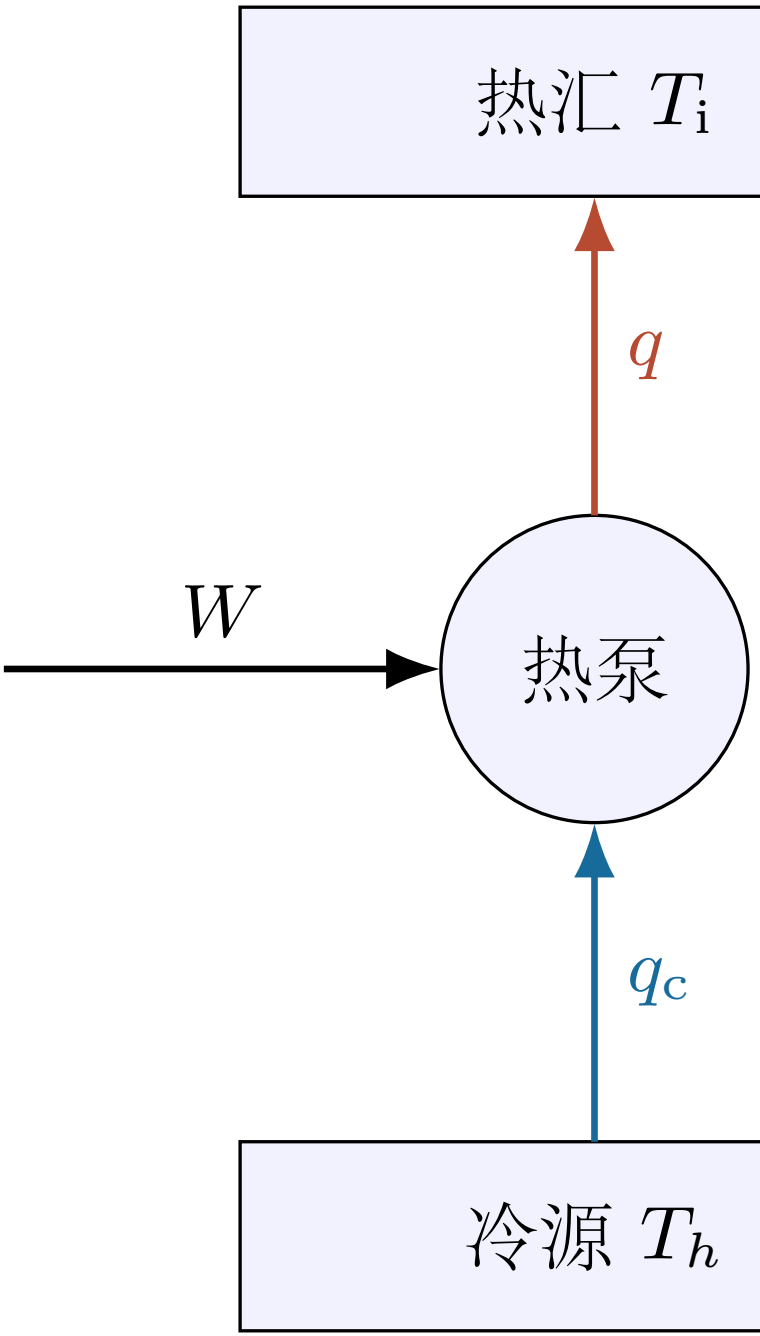


$Q = 0$，流出气体携带焓

升空时质量固定



$$q + P_\mathrm{sun} = P_\mathrm{loss}$$



$$q = q_{\mathrm{c}} + W$$

$$\frac{q}{T_{\mathrm{i}}} \geq \frac{q_{\mathrm{c}}}{T_h}$$